

Encoder Decoder Code

```
message = input("Enter a message to encode or decode: ") # Get a message
```

```
message = message.upper()    # Make it all UPPERCASE :)
```

```
output = ""                  # Create an empty string to hold output
```

```
for letter in message:      # Loop through each letter of the message
```

```
    if letter.isupper():    # If the letter is in the alphabet (A-Z),
```

```
        value = ord(letter) + 13 # shift the letter value up by 13,
```

```
        letter = chr(value)    # turn the value back into a letter,
```

```
    if not letter.isupper(): # and check to see if we shifted too far
```

```
        value -= 26          # If we did, wrap it back around Z->A
```

```
        letter = chr(value)  # by subtracting 26 from the letter value
```

```
    output += letter         # Add the letter to our output string
```

```
print("Output message: ", output) # Output our coded/decoded message
```